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LP6 GUITAR SCALES SYSTEM: a recipe to get quickly acquainted with the system.

Best to read through the PDF to gain an awareness of the bits and pieces. Then focus on using one key for learning: Am or C is good (same scale notes but different degrees).

1. recall or look up the Origin # for your major key (or the relative major of your minor key); here it's C.
2. calculate or visualize the first row or two, of pattern numbers and expand as you play further down the neck. (pages 13, 14, in the PDF)
3. check mentally if any row is one of the alignment anomalies ('175' , '431', or '764'). This becomes second nature after a bit.
4. check if your key (E, F, or B) demands a shifted Division (page 15, 16 ...)

The power and beauty of this system lies in the fact that <u>everything</u> you learn in the basic "Major scales and modes" system, is used to extend the system to PENTATONICS, CHORD NOTES in scale pattern, HARMONIC MINOR scale, and virtually any other scale. The way this system is 'extended', is by taking the patterns learned in the base system and applying a set of 7 'DESCRIPTORS' to the 7 six-note patterns, to reveal a sub-pattern or alter the pattern as desired. For example, … if you wish to have a scale that is like the natural minor scale but has the 5th degree note sharped, you would create a set of 7 Descriptors, as shown at the end of this document.

Background

Music scales are the "alphabet" of the music language. With this alphabet we can learn to "spell" musical "words" such as chords, and "licks". We can then assemble some of those "words" into "phrases", … then "sentences", … and eventually into "stories" which appeal to many. This is possible because, as in a word based language where we all recognize familiar words and phrases, and are mesmerized by their eloquent use, music components like chords and licks must be "spelled" correctly or they not harmonious. "Words" must be put in the right order, or a musical idea does not fit the melody … and so on. When all is right, the musical "story" can be magical.

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Most modern guitar music is written using a 7 note scale called the Diatonic Scale. The Diatonic Major Scale with its familiar sounding "DO-RE-MI-FA-SO-LA-TI-DO", is used to create and play the majority of guitar music today. This is so, because all of the Minor Scale and all Mode Scales, as well as the Pentatonic scales are derived from the Major Scale. There are 144 scales you can play, just by knowing the Major scale; ([7 Major modes + 5 Pentatonic modes] * 12 keys = 144).

The consecutive notes or tones of the Major Diatonic Scale are separated by a full tone or semi-tone. The separation is usually referred to as an 'Interval'. A full-tone is equal to 2 semi-tones. The major scale always has a semi-tone between Degrees 3 & 4, as well as between 7 & 1.

A semi-tone interval is one fret space on the guitar. The "C" and "E" major scale notes and intervals can be represented as shown below.

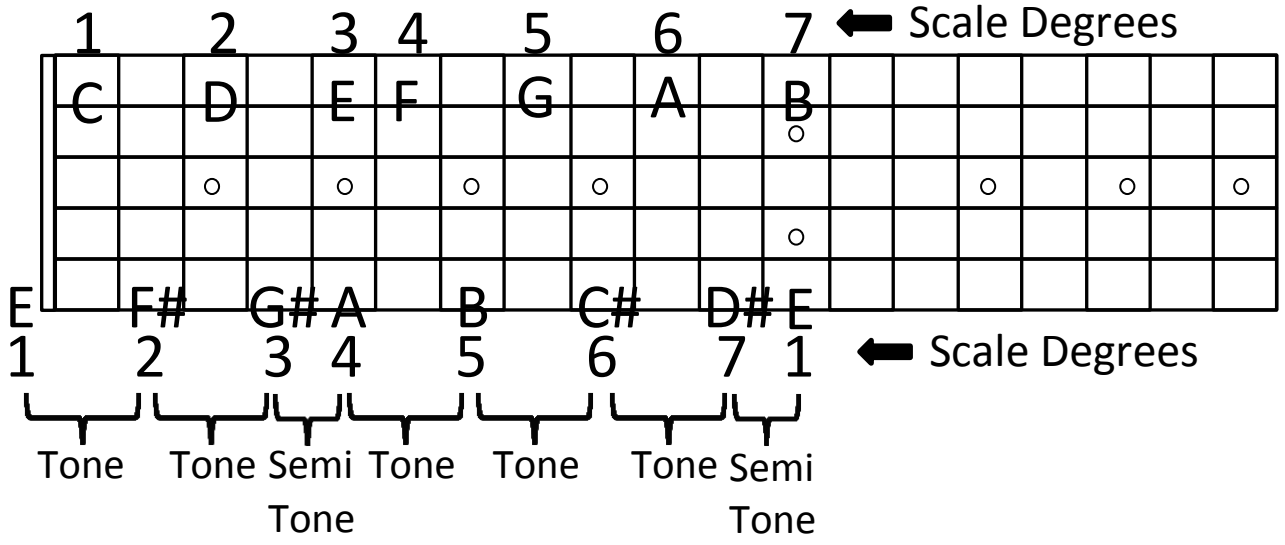

Figure1

Combined with the number of different scales and modes, each having 12 keys, learning to play a variety of different music on the guitar requires the knowledge of many scales and is a formidable challenge.

The method here, can be used for other stringed instruments such as bass guitar, double bass, viol, some mandolins and any other instrument tuned in fourths which have fretboards or fingerboards without frets.

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Summary

This invention provides a method and visualization, to partition the large problem of learning many scale patterns, into:

- A single system that works coherently for 144 scales
- Memorizing 7 small, 6-note Patterns, which are numbered
- Methods for locating and placing "Pattern Numbers" over the entire fretboard ("Origin and "Placing")
- A method for placing a Pattern at each "Pattern Number" location (Divisions)
- A method for aligning the notes within the Patterns (alignment)

The Patterns are numbered in such a way that a set of 12 patterns covering the entire fretboard, has a very simple mathematical relationship between its Pattern numbers. See Figure 2 and Figure 3 below.

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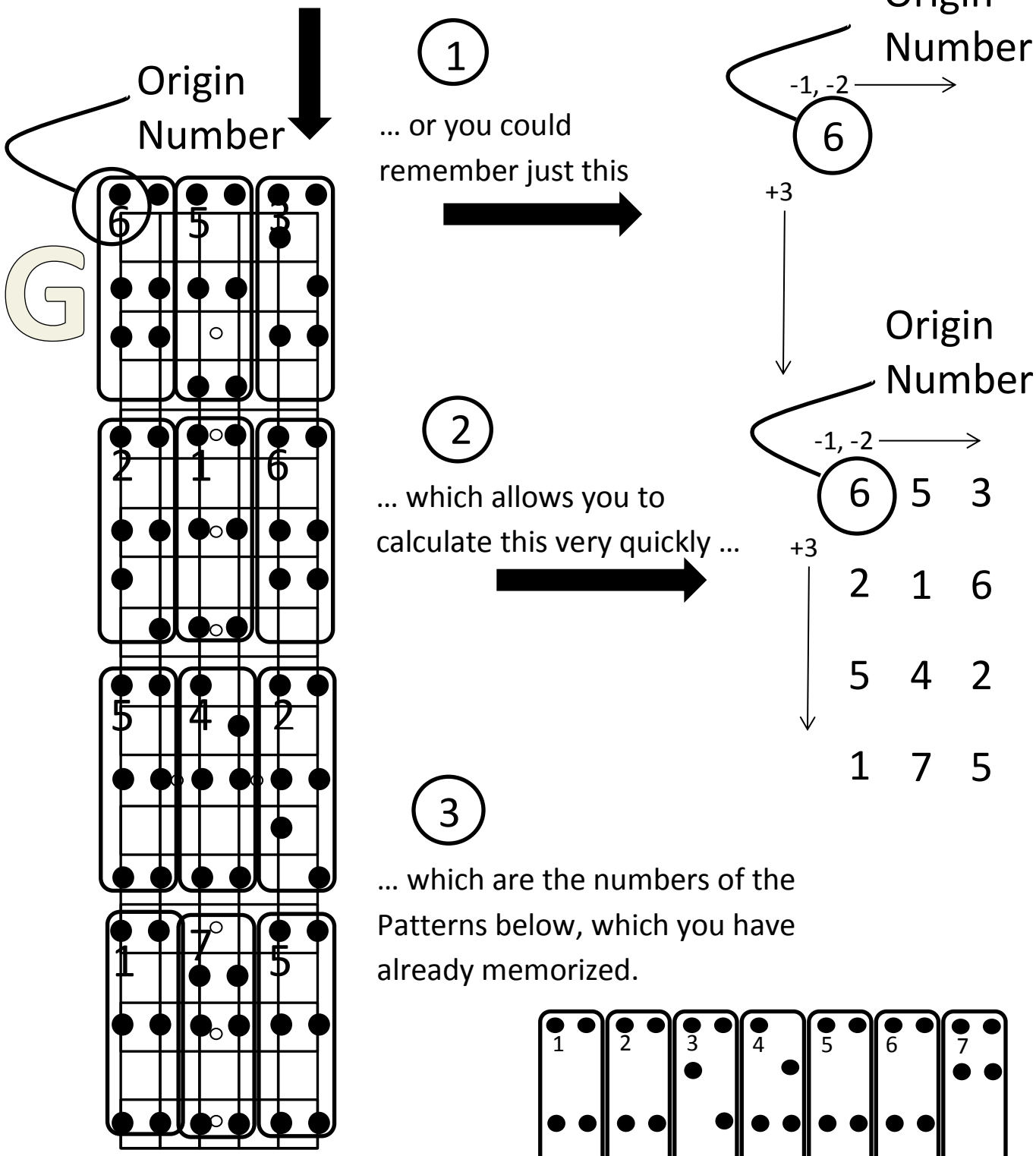
All the scale notes of G major are shown below on a guitar fretboard. It would take quite some time to memorize the dot pattern (72 notes of G scale) over the entire fretboard.



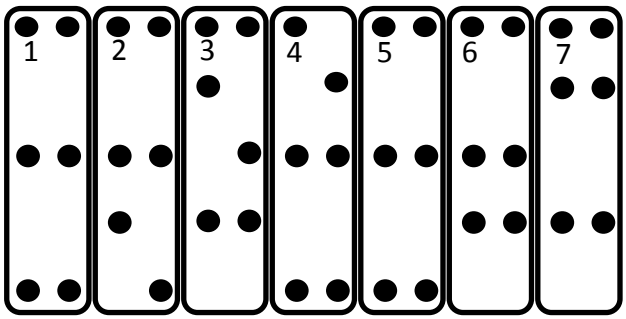


Figure 2 … continued on Figure 3

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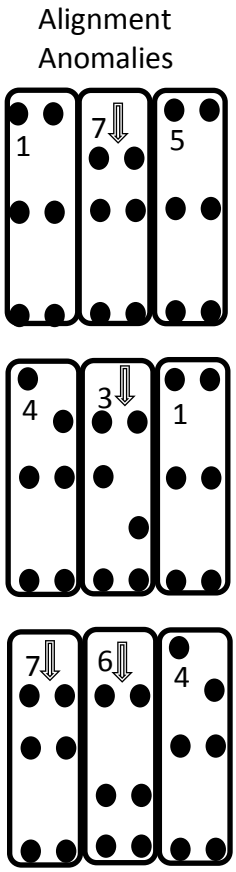
… continued from Figure 2

GENERAL DIVISION OF THE FRETBOARD

(for all scale patterns except : E, F, B, and sharp keys)

4 Recall if any Divisions are shifted (based on the key) …

5 Check if any row calculated in step 2, is one of the alignment anomalies …



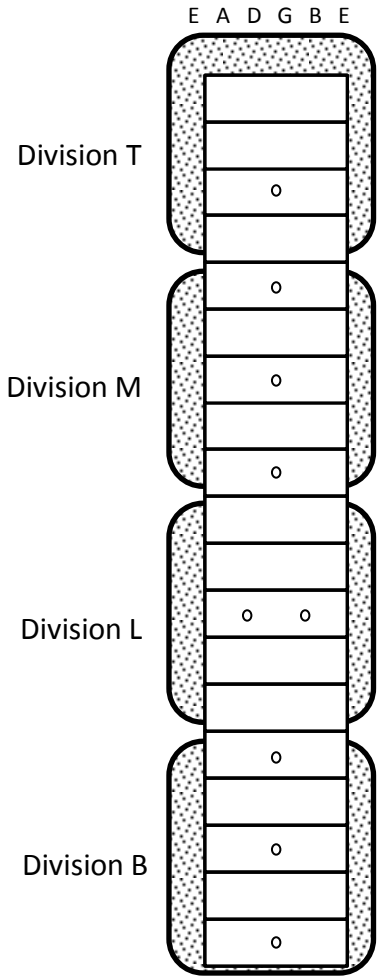


Figure 3

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Detailed Description

Sub-patterns

Because the guitar is tuned with Perfect 4th intervals between adjacent strings (except G - B), we can create a sub-pattern on any string that also spans a Perfect 4th, and it becomes possible to count sequentially, the scale degrees, by continuing to the next sub-pattern vertically or horizontally on the fretboard , which suggests that 2 groups of 3 notes side-by-side (2 sub-patterns) will create an elegantly simple sequence of 'scale degree' numbers across adjacent groups of 6 notes. See Figure 4 and Figure 2.

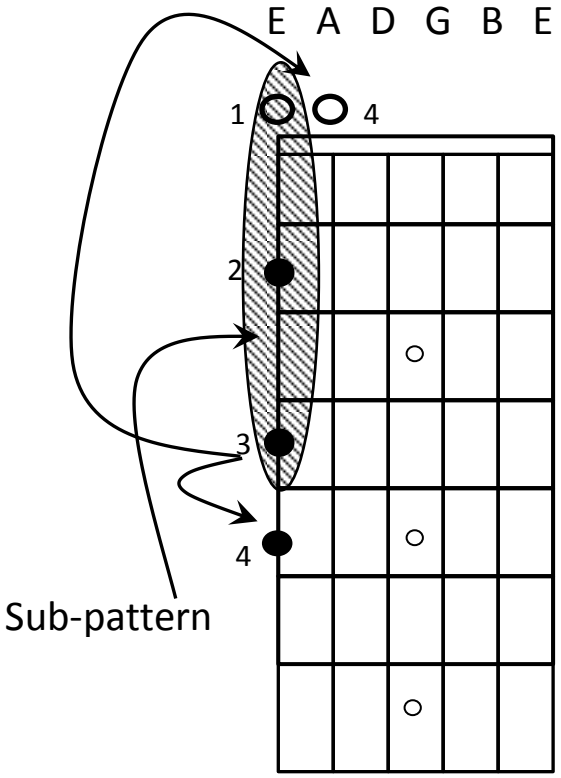

Figure 4

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Sub-patterns shown in Figure 5, are created to represent the Major scale intervals for 3 consecutive notes starting with each of the scale's degrees such that each represents a total interval of a perfect 4th which is equal to the interval between the strings of a guitar.

Sub-Patterns

… for diatonic major scales.

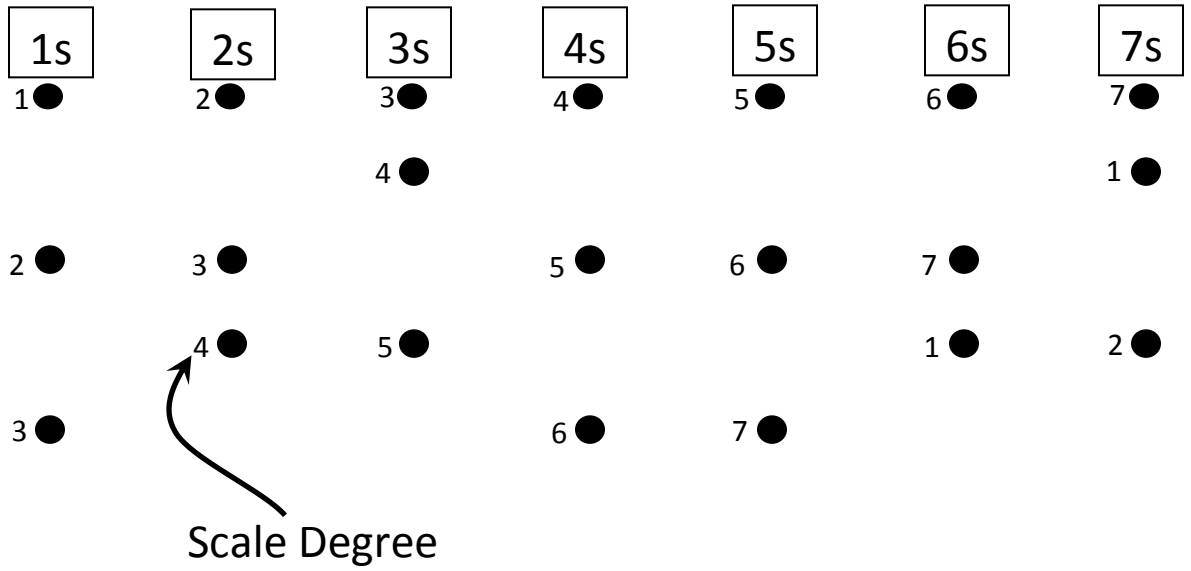

Figure 5

6-Note Patterns

Taking 2 sub-patterns that contain scale degrees that are sequential, we can build seven 6-note Patterns as shown in Figure 6, similar to the 7 sub-patterns of 3 notes.

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Patterns

… for diatonic major scales.

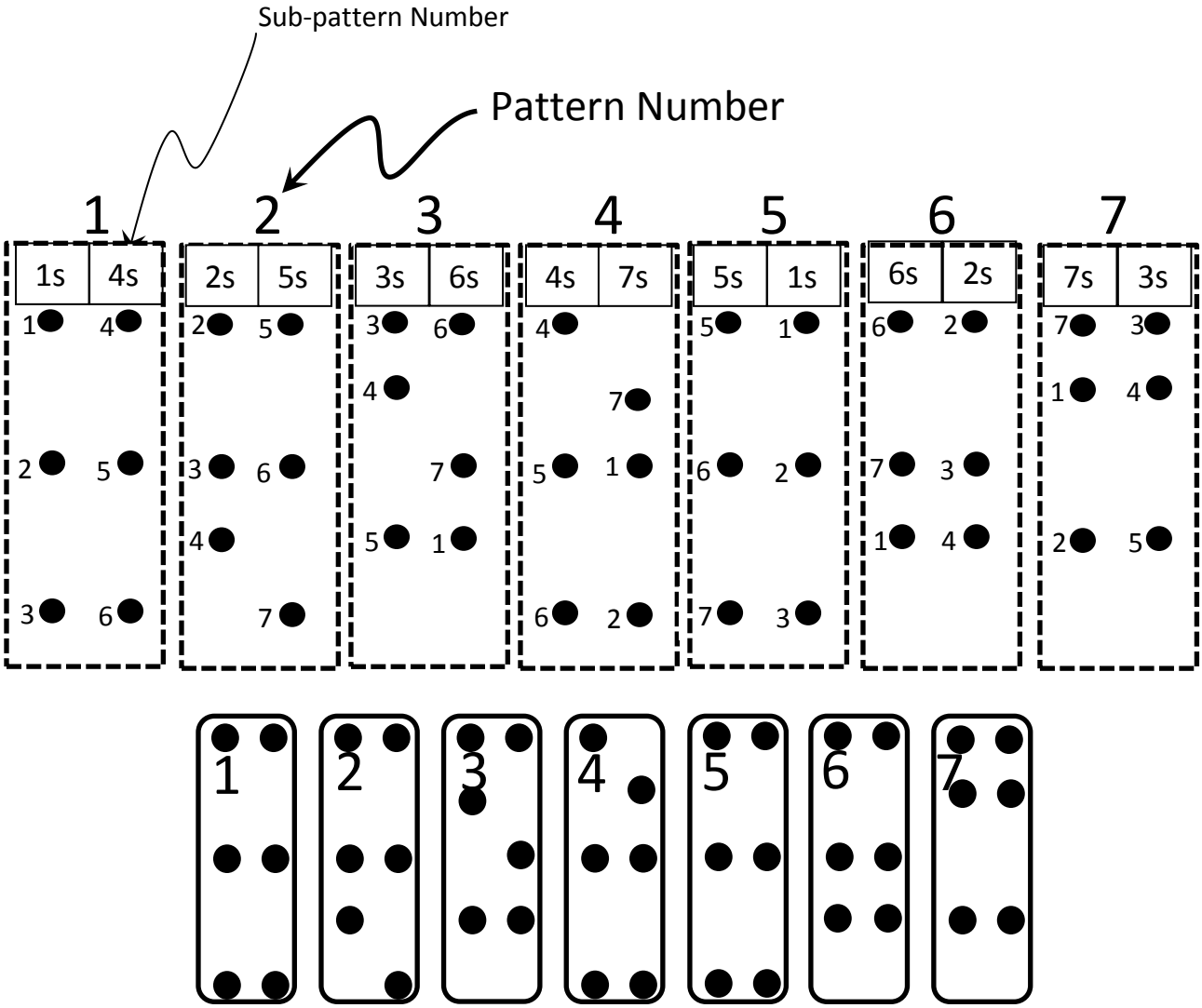

Figure 6

Note that the scale degree numbering within a Pattern is coherent with the numbering of the sub-patterns. Example: Assuming you know the sub-patterns … If you are trying to remember the second sub-pattern of the #2 Pattern, you can simply count 2, 3, 4 which leads to 5. So sub-pattern #5 is the 2nd half of Pattern #2.

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Patterns are very easy to memorize …. There are really only 4 patterns to memorize …

Patterns 1 and 5 are identical

Pattern 2 is pattern 4 inverted

Pattern 6 is pattern 7 inverted

Pattern 3 second sub-pattern is the first sub-pattern inverted

Figure 7 shows the 6-note Patterns for the scale of “C” major, over the entire fretboard.

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Example: “C” major scale

Note: all patterns start at the top of each Division.

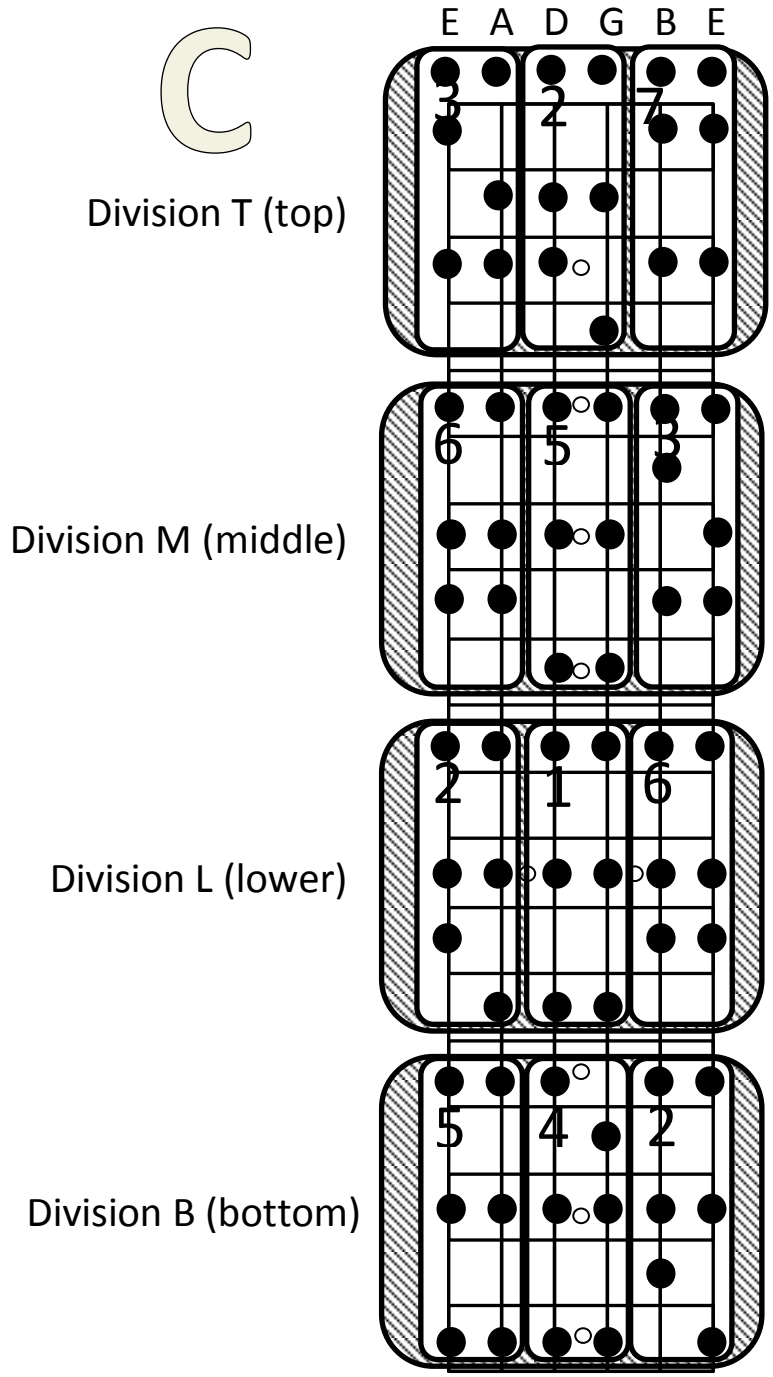

Figure 7

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"Origin Number"

The purpose of this system is to create an 'easy to remember' order to the patterns, … so a convenient "origin" or "anchor" point must be well chosen.

The "origin" for this system is at the upper leftmost position of the fretboard … ie at the open low "E" note. It is the anchor for the entire matrix of Pattern Numbers.

Having chosen the low "E" as our "origin" we now must assign the major scale of "E" an "Origin Number" of "1", so that Pattern 1 with the 1st degree of the scale (first dot # of the pattern) falls on the low "E" location (ie open low "E" string). See Figure 1 and Figure 8.

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Example: “E” major scale

Note: Division “B” is lowered one fret as per Figure 15. With that done, all patterns start at the top of their division, except the 3 anomalies, 175, 431, & 764, as described on page 22.

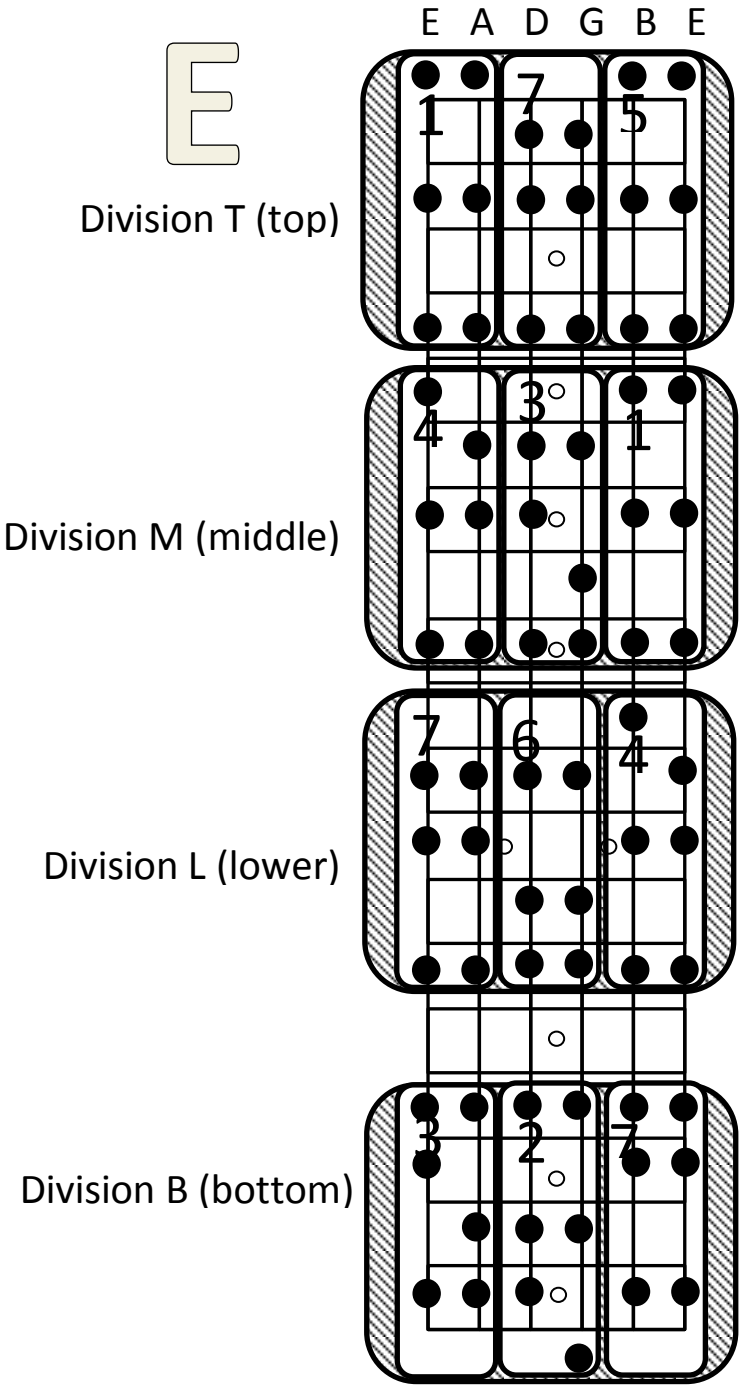

Figure 8

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The "Origin Number" for any of the other keys of the major scale is found as follows:

- Write down the notes of the key, in order, all in a row, starting with the root note.
- Write the scale degree numbers in a row below
- Look for the "E" note in the row of notes, and see what scale degree number is below it. That is the "Origin Number" for your key. See Figure 9 for an example.

Example: What is the "Origin Number" for the key of "B" ?

Scale note	B	C#	D#	E	F#	G#	A#
degree	1	2	3	4	5	6	7

Find the Scale note "E" …. Look at it's Degree number in the box below ….

Answer: "Origin Number" for "B" is 4.

See Figure 10 for the "Origin Number" for all keys of the major scale

Figure 9

For the Major Scale, the "Origin Number" is assigned to the 12 keys as follows:

Key	F, F#	G, G#	A, A#	B	C, C#	D, D#	E
Origin #	7	6	5	4	3	2	1

*** All sharp keys use the same "Origin Number" as the natural key, but all the Divisions are simply shifted one fret down. (See 'DIVISIONS' in Figures 13, 14, 15, 16, 17, and 18)

Figure 10

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Generating the "Pattern Numbers" Array

It turns out that the numerical sequence between adjacent groups of 6-note Patterns, is so simple that you can map the entire fretboard for a given key, almost instantly.

The sequence Numbers are composed of the 7 degrees of the diatonic scale: 1, 2, 3, 4 , 5 , 6, 7.

You need to be able to count through these 7 numbers, forward, or backward by 1, 2, or 3s. Example: to count through the scale degrees by 3s …. 1, 4, 7, 3, 6, 2, 5, 1 ….

Note that to count beyond 7, you must start over at 1, as though the 7 numbers were in a circle. See figure 11.

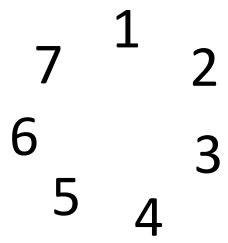

Figure11

If the instrument was tuned entirely in "Perfect 4th" intervals between the strings, the sequence of "6-note pattern" numbers going across the fretboard would be for example : 4 3 2, or 7 6 5, or 1 7 6 … in other words, counting down in decrements of 1.

However the Standard tuning for a guitar has a Major 3rd between G and B strings, so the sequence going across the fretboard from low E to high E will be for example 431, or 764, or 175.

The sequence vertically down the fretboard toward the guitar body, is created by counting up in threes.

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See Figure 12 for an example of Pattern Numbers for the key of “E” over the entire fretboard.

Pattern Numbers for the key of “E” over the entire fretboard

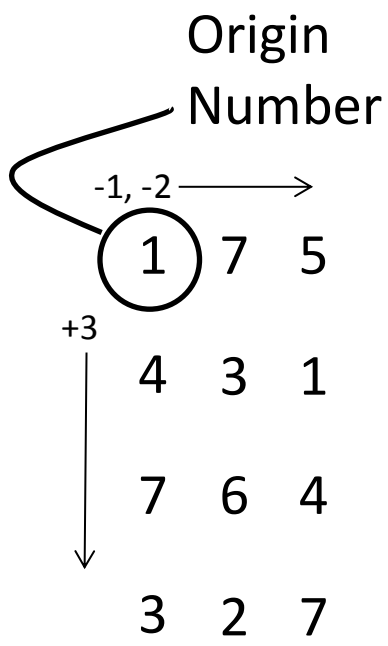

Figure 12

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Placing Patterns on Fretboard (Divisions)

Each division holds one horizontal row of 3 patterns.

To place all patterns at their matching “Pattern Numbers” on the fretboard, we must partition the fretboard using rectangles called Divisions which are key dependent. The Divisions reduce dot alignment exceptions in some Patterns. The coherency and simplicity of this invention is made possible by these Divisions. They reduce complexity because they are key based rather than Pattern based, so depending on your choice of key, there may be no said exceptions at all.

There are 6 fretboard layouts for Divisions, depending on the desired key of the scale:

1. A “General” layout for all natural keys except E, F, and B; - see Figure 13.
2. A “General” layout for all sharp keys except F#; - see Figure 14.
3. A specific layout for the key of “E” ; - see Figure 15.
4. A specific layout for the key of “F” ; - see Figure 16.
5. A specific layout for the key of “F#”; - see Figure 17.
6. A specific layout for the key of “B” ; - see Figure 18.

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GENERAL DIVISION OF THE FRETBOARD

(for all key scale patterns except : E, F, B, and sharp keys)

Placing patterns in Divisions like those to the right, rather than directly on the fretboard, is what makes the coherency and simplicity of the LP6 System possible. (In this particular diagram (Figure 13), it is equivalent to placing patterns directly on the fretboard). See Figures 14, 15,16, 17, & 18, where it is *not* equivalent.

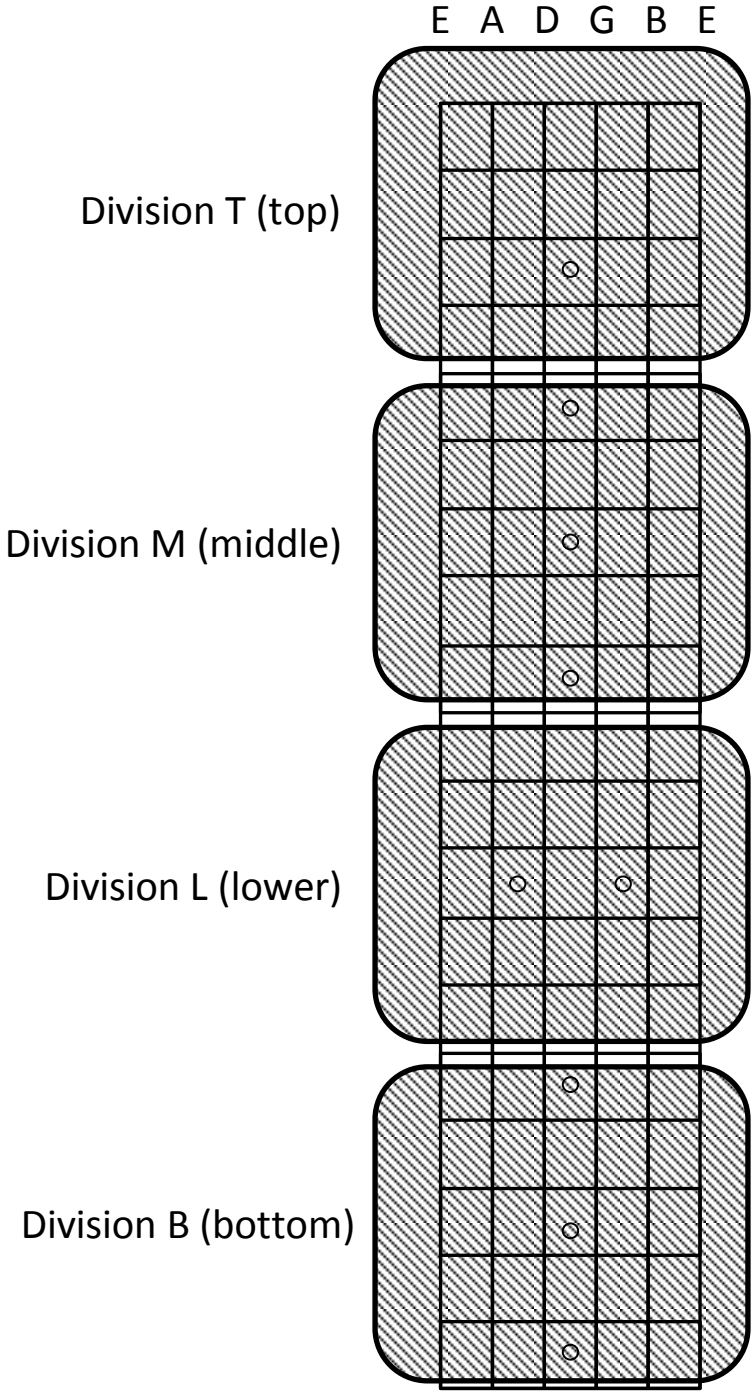

Figure 13

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GENERAL DIVISION OF THE FRETBOARD

(for all sharp scale patterns except : F#)

Drop all Divisions one fret.

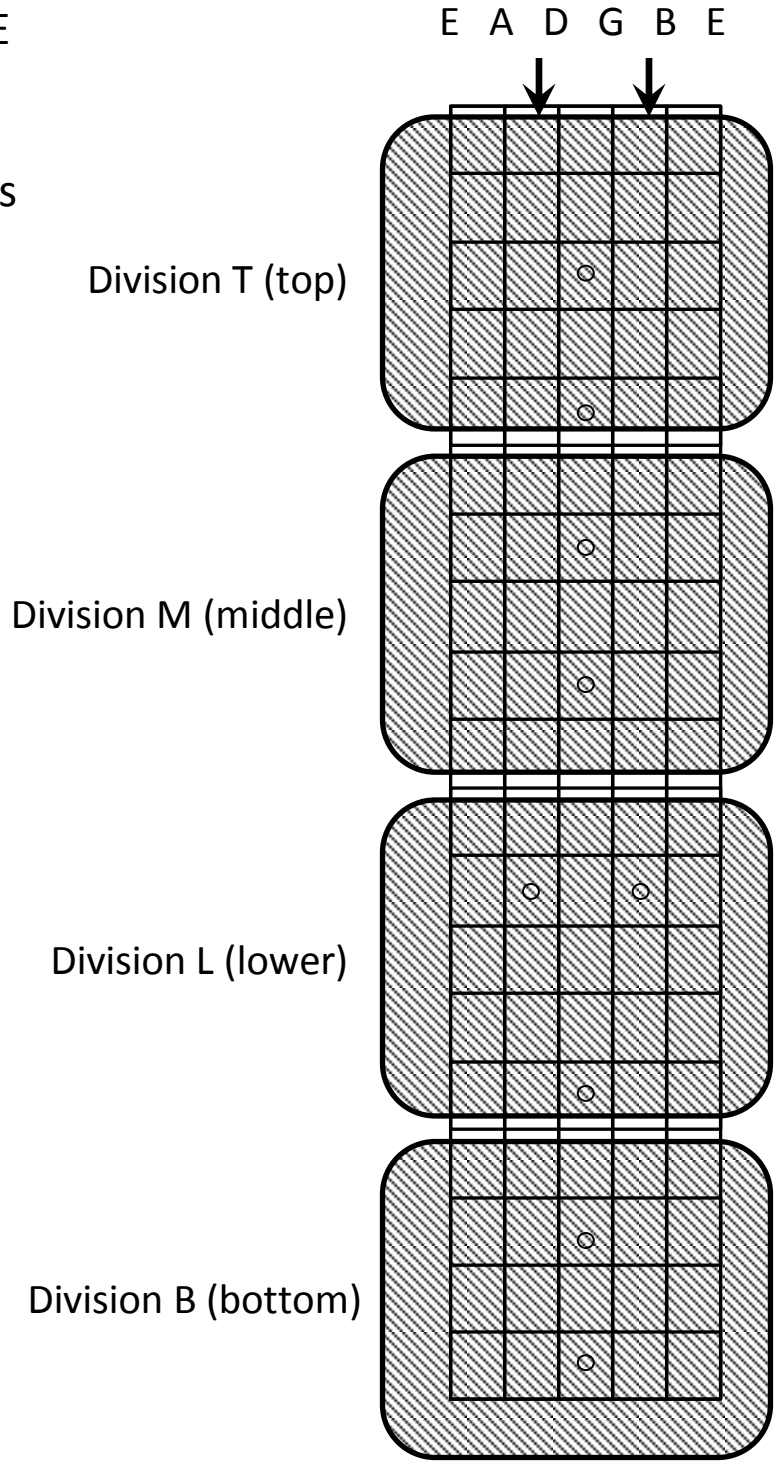

Figure 14

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DIVISION OF THE FRETBOARD
for “E” SCALE PATTERN

Division T (top)

Drop Division “B” one fret.

Division M (middle)

Division L (lower)

Division B (bottom)

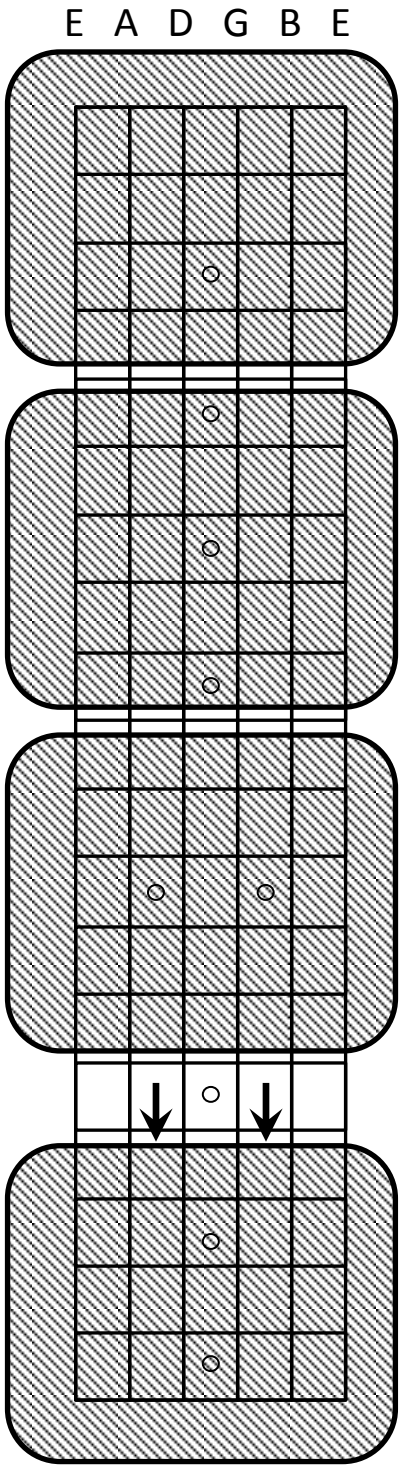

Figure 15

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DIVISION OF THE FRETBOARD
for “F” SCALE PATTERN

Raise Division “T” one fret.

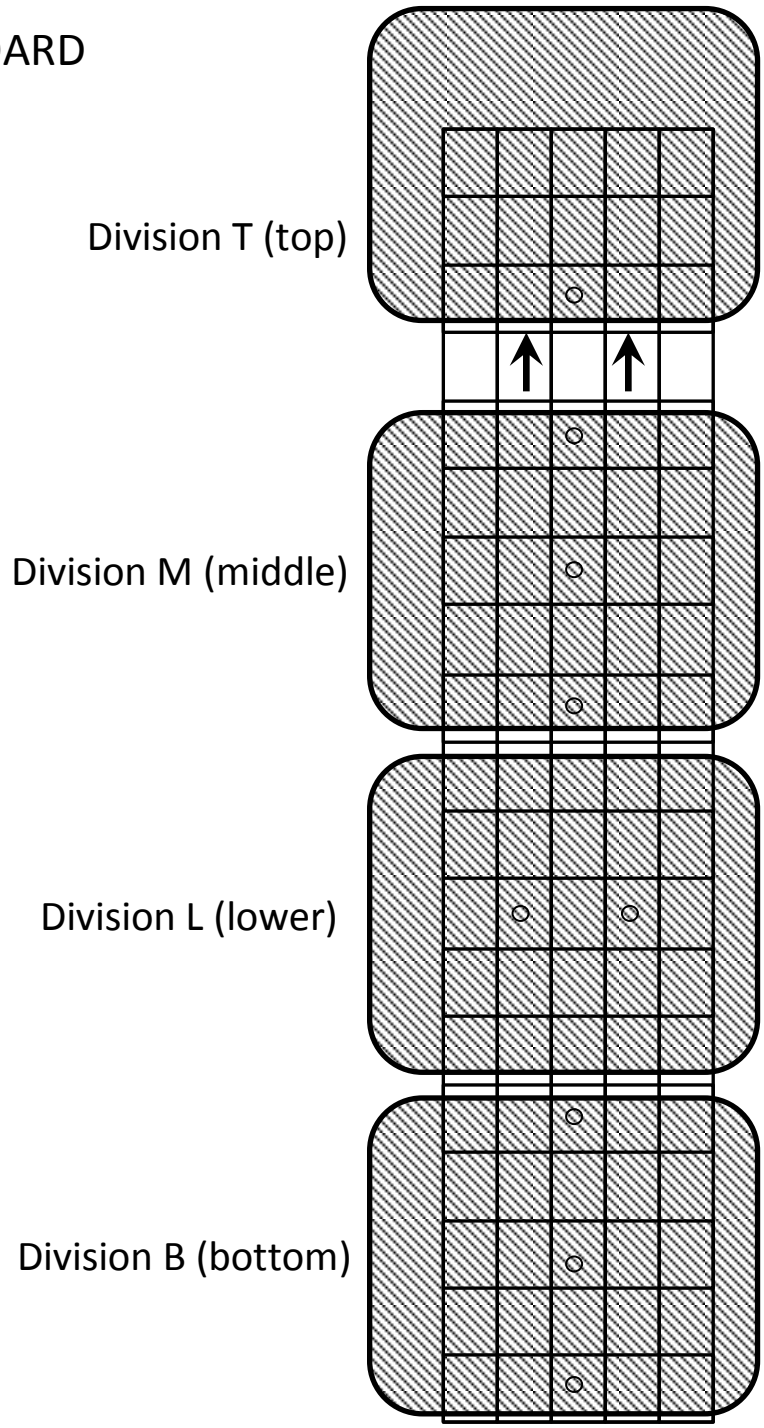

Figure 16

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DIVISION OF THE FRETBOARD for “F#” SCALE PATTERN

Lower Divisions “M”, “L”, & “B” one fret.

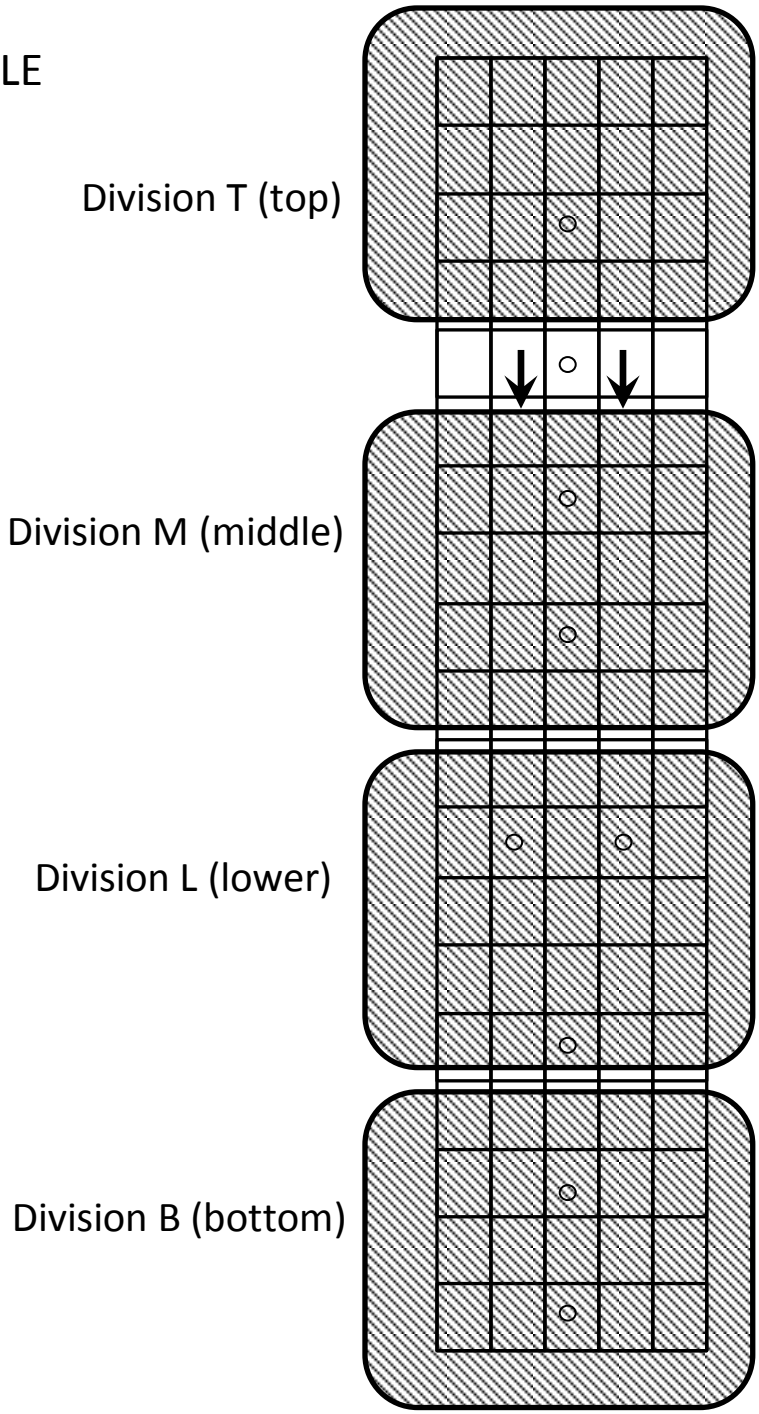

Figure 17

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DIVISION OF THE FRETBOARD for “B” SCALE PATTERN

Division T (top)

Division M (middle)

Drop Divisions “L” & “B” one fret.

Division L (lower)

Division B (bottom)

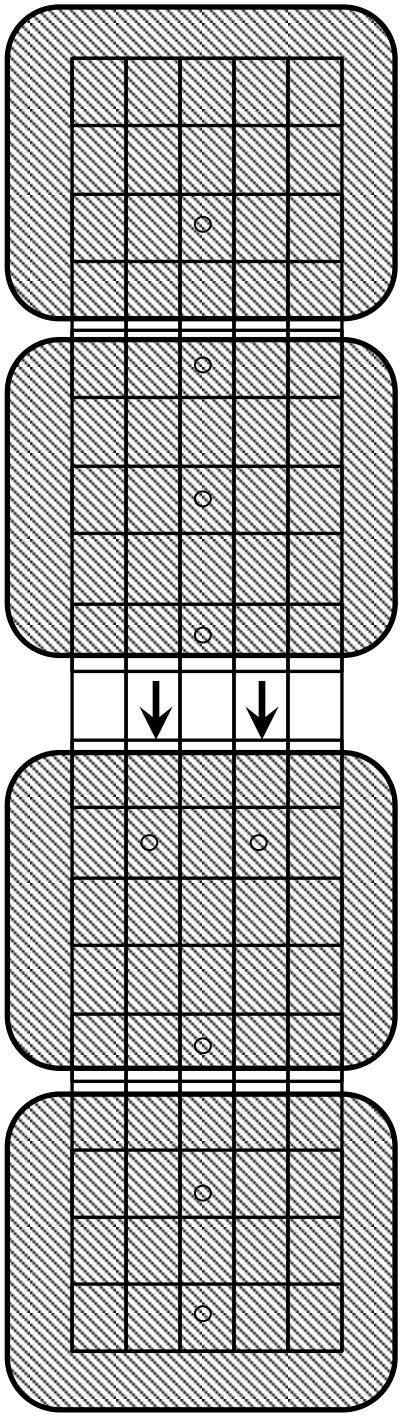

Figure 18

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Alignment of dots (notes) with Patterns

Because the 6-note patterns are 5 frets in size, and some sub-patterns are 4 frets in size, a method of aligning the 4 fret sub-patterns within a 6-note Pattern with 2 said sub-patterns, is necessary.

The method is extremely simple:

- Align all sub-patterns at the top of each 6-note Pattern except for 3 anomalies
- The Anomalies occur only in 3 of the row sequences of the Patterns: 175, 431, 746. The anomalous sub-patterns are aligned 1 fret down from the top of the 6-note Pattern.

Example in Figure 2, G scale: All patterns start at the top of each pattern group except the 175 group at the bottom of the fretboard, where the “7” pattern starts one fret down.

The 3 such row pattern groups that contain a pattern (or 2) that doesn’t align at the top of the pattern box are : 175, 431, and 764.

In 175, the 7 pattern is one fret lower.

In 431, the 3 pattern is lower.

In 764, the 7 & 6 patterns are lower.

See Figure 19.

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Alignment Anomalies

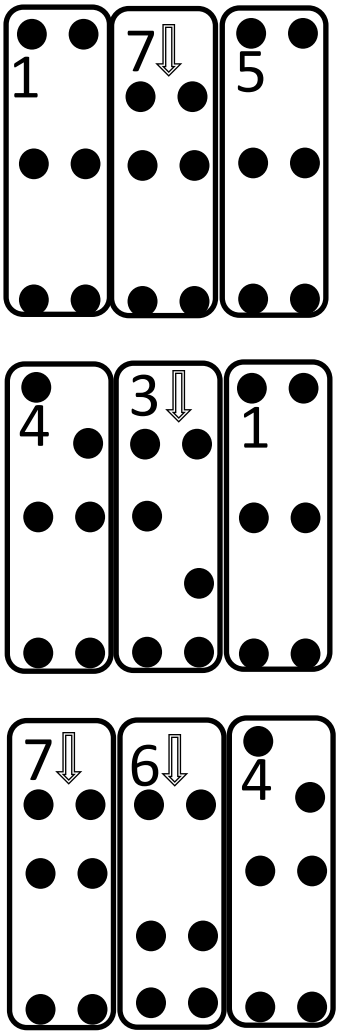

Figure 19

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Example: “F” major scale

Note: Division “T” is raised one fret as per Figure 16. With that done, all patterns start at the top of their division, except one anomaly, 764, as described in Figure 19 .

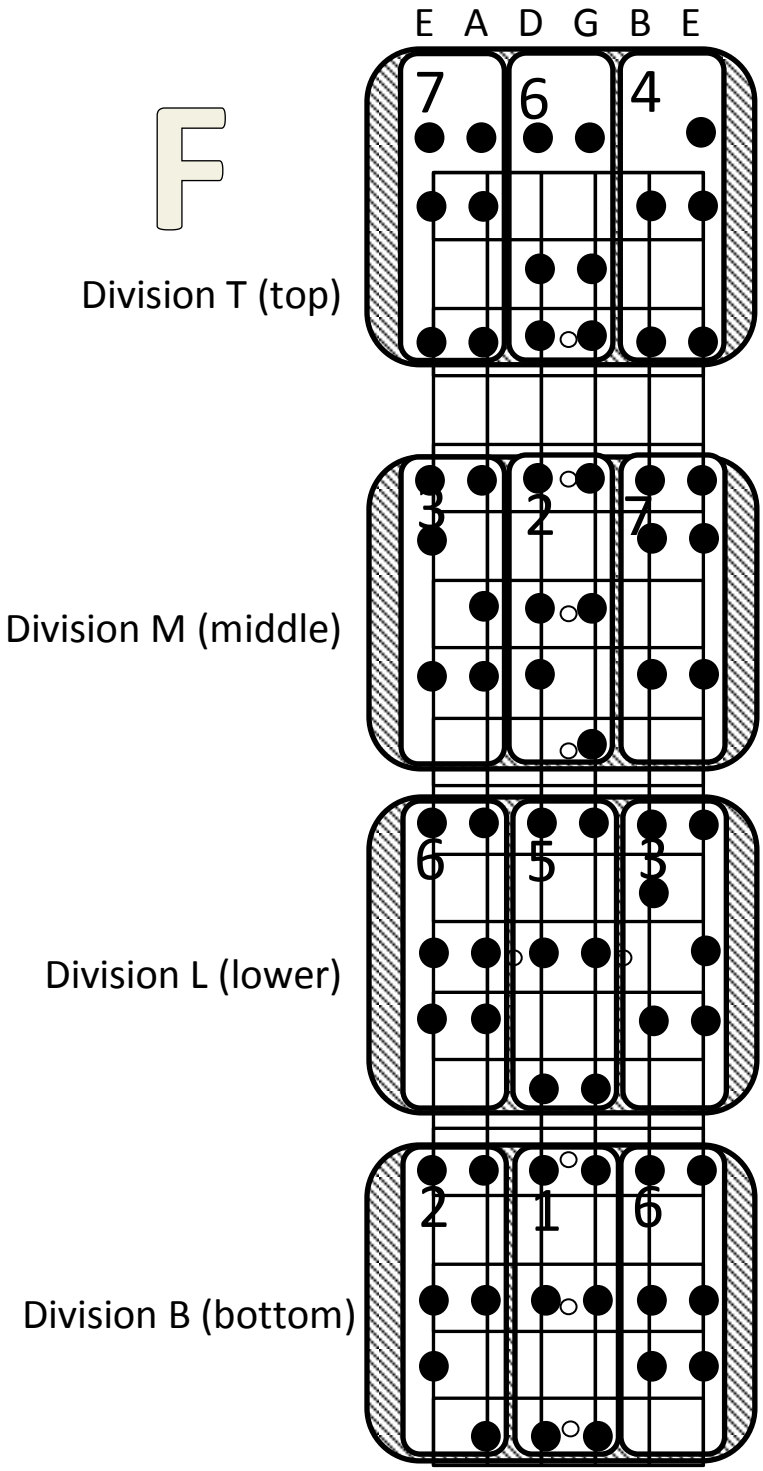

Figure 20

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Finding the proper scale for major scale keys

- Choose your key
- Lookup the "Origin Number" in Figure 10 or from memory
- Calculate or visualize pattern numbers and place them on the fretboard according to the system described above

Finding the proper scale for Modes of the major scale

- Choose your key, and Mode number.
    - (Ionian=1, Dorian=2, Phrygian=3, Lydian=4, Mixolydian=5, Aeolian=6, Locrian=7)
- Find the interval distance between your Mode number and 1 (the root).
    - Example for "E Aeolian" : Mode is 6, so the interval distance from Degree 6 to Degree 1, of the E major scale, counting forward, is +1.5 tones, or counting backward, is -4.5 tones.
- Find corresponding key in major scale.
    - From the example above: "E" + 1.5 tones on the "E major scale takes us to the "G" note. So the "E" Aeolian has all the same notes as G major scale except its Degree numbers are shifted back by Mode # -1; in our example 6-1=5; or forward by 7-5=2; … for "E" the root is at the Mode # (=6 for Aeolian) ; so … E degree # = G degree # + 2.
- Go to the "Finding the proper scale for major scale keys" section and follow the steps using the "corresponding" key found in the previous step

Finding the proper scale for the Natural Minor scale

- Follow the procedure for the "Modes of the major scale" in the preceeding section, for the Aeolian mode (6th mode).

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Finding the proper scale for the Minor Pentatonic scale

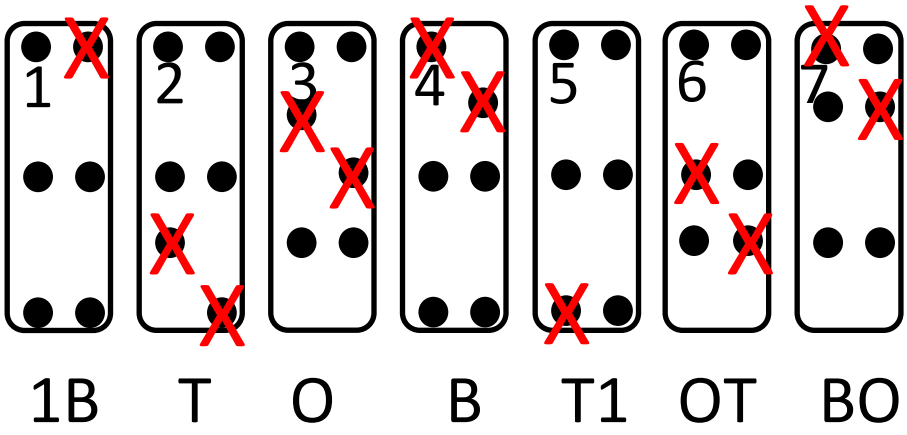
The Pentatonic scales are 5 note scales which are the same as the 7 note scales (Diatonic), with 2 notes omitted. The Minor Pentatonic scale is the same as the Dorian, Phrygian, and Aeolian Modes (minor modes) with the 2 and 6 notes omitted, or the 3 & 7, 1 & 4, and 4 & 7 notes respectively, of the corresponding major scale keys.

- Choose your Minor Pentatonic key
- Choose the same key in one of the following Modes: Dorian, Phrygian, Aeolian
- Find the corresponding major scale key for the chosen Mode above
    - Example: "A" in Minor Pentatonic = "A" Aeolian, which is the same notes as the corresponding "C" major scale with the "A" Aeolian root found at the 6th degree of the said "C" scale.
    - Omit notes of Degree 4 & 7 of the "C" corresponding scale (= 2 & 6 of the Aeolian Mode)
- Go to the "Finding the proper scale for major scale keys" section and follow the steps using the key found in the third step above

NOTE: ***
An easy way to remember which **notes of the Diatonic Scale** (such as A minor or A Aeolian) are the **notes of the Minor Pentatonic** scale (with 4&7 notes omitted) is as follows:

Pentatonic notes – Descriptor shapes for minor modes

1B means all notes of the left & bottom notes of the right column; T means the top of both columns; O means the outer notes only; B means the bottom notes of both columns; T1 means top notes of left and all notes of right column; OT means outer notes of left and top notes of right column; BO means bottom notes of left and outer notes of right column.
Notice that Pattern 5 is the invert of Pattern 1, (1B & T1); same for 6 & 7 (OT & BO), as well as 2 & 4 (T & B).



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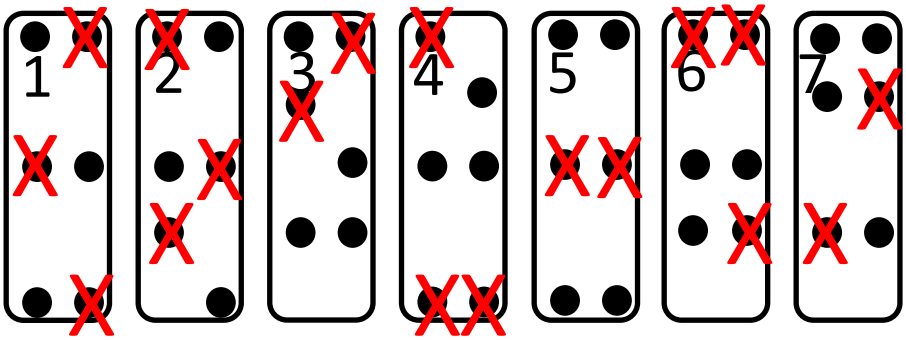
Chord notes - Descriptor shapes for MODES.

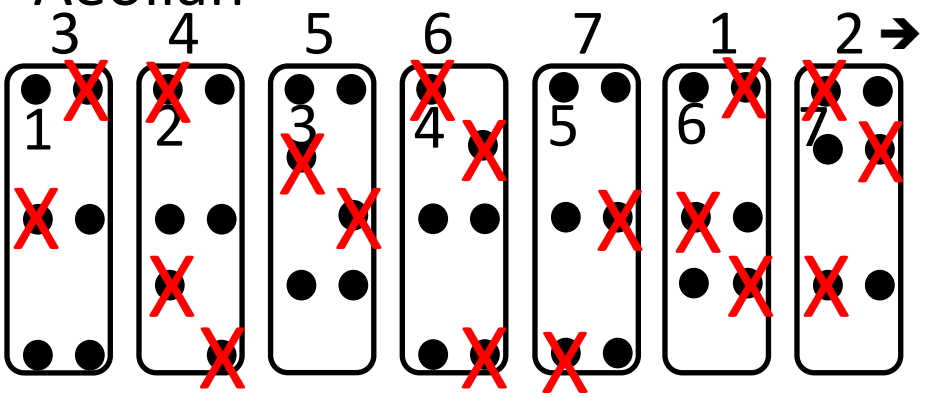
If you wish to visualize the chord tones within your scale patterns you can memorize a 'Descriptor' shape that roughly but uniquely describes the shape that the chord tone dots make within a pattern box. You may find better ones ...

Major key or Ionian



D C ≥ -‘ O r r.

Aeolian



➔ Translated pattern #

≥ -‘ O r r. D C

Translated pattern # of mode

1	2	3	4	5	6	7
D	C	≥	-‘	O	r	r.

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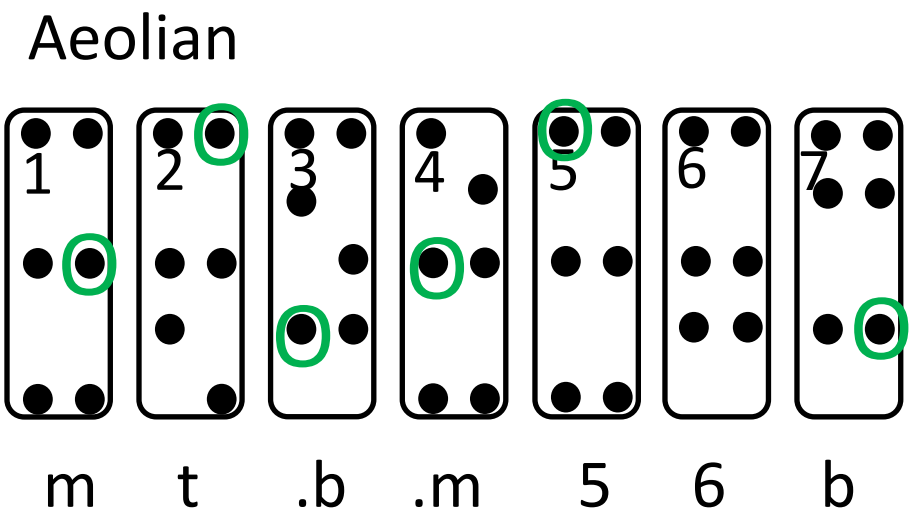
5th degree notes – Descriptors for Aeolian mode scale

Sharp these 5th notes to get the Harmonic Minor scale.

To extend this system to the Harmonic Minor scale, we need a set of 'Descriptors' that reveal the location of the 5th degree note (of the Aeolian scale) within each pattern box. Once you know it's location is, you can simply sharp that note to get the Harmonic Minor scale.

So in pattern 1 box, the 5th degree note is located in the middle of the 2nd column, so we use "m"; Then in boxes 2 through 7, the 5th degree moves up, and then 'snakes' over to the 1st column and again moves up. So, remember the 'snake' because it helps you know the position of the 5th in the next pattern box. If the position of the 5th degree note is in the first column, i put a period in front of the position to show that, as in ".m" for the 4th pattern box.